

Building a Simple Neural Operator Experiment

So far the argument has been made mostly in symbols. We said that a neural operator learns a map between *function spaces*, and the example we used was the integration operator

$$\mathcal{I}[f](x) = \int_0^x f(s) ds, \quad \text{that is,} \quad f(x) \xrightarrow{\mathcal{I}} u(x).$$

You hand $\mathcal{I}$ an entire function f and it hands back an entire function u . Change f and you get a different u . Nothing here is a number being mapped to a number; the *input itself is a curve*.

That example is clean because we can write $\mathcal{I}$ down in closed form. The interesting claim is that the same picture applies when the operator is something we *cannot* write down — for instance, the map from a boundary condition to the solution of the Navier–Stokes equations. That is what this experiment builds:

$$g(x) \xrightarrow{\mathcal{G}} [u(x, y), v(x, y)].$$

Here $g(x)$ is the horizontal velocity profile along the moving top lid of a 2D lid-driven cavity, $\mathcal{G}$ is the steady Navier–Stokes solution operator at fixed nondimensional viscosity coefficient, and $[u, v]$ is the resulting interior velocity field. We will train a network $\mathcal{G}_\theta$ to approximate $\mathcal{G}$, and then feed it lid profiles it has never seen.

The coefficient in the nondimensional equations is held fixed at $\text{Re} = 100$, using the reference scales $U_{\text{ref}} = L = 1$, so every sample uses the same viscosity $\nu = 0.01$. What changes is the top boundary function. Because its amplitude also changes, a Reynolds scale formed from each case’s actual lid speed is not constant; we quantify that range below.

This is an operator-form learning experiment, but a deliberately modest one. The lid functions come from a four-coefficient family. DeepONet receives each input as 41 function samples rather than those four coefficients, yet the family’s intrinsic dimension is still four. The experiment demonstrates the branch/trunk function-to-field construction; it does not show that such a model outperforms a network trained directly on the coefficients.

Everything below — every number, every figure — comes from code in this repository that was actually executed. Nothing is illustrative.

Why DeepONet?

There are several neural-operator architectures (Fourier Neural Operators, graph kernel operators, and others). We use **DeepONet** here for a pedagogical reason: its structure makes the function-to-function idea legible.

DeepONet splits the problem into two networks that answer two different questions:

- The **branch** network reads the *whole input function*, sampled at m fixed sensor locations: $[g(x_1), g(x_2), \dots, g(x_m)]$. It answers “*which problem is this?*”
- The **trunk** network reads a *query coordinate* (x, y) . It answers “*where in the domain are we?*”

They meet in a dot product:

$$u(x, y) \approx \sum_{k=1}^p b_k(g) t_k(x, y) + b_0.$$

Read that sum carefully, because it is the whole idea. The trunk outputs $t_k(x, y)$ form a **learned basis** of functions over the cavity. The branch outputs $b_k(g)$ are the **coefficients** that this particular lid profile puts on that basis. Feeding in a new g does not change the basis — it changes the coefficients, and therefore the reconstructed field.

This is a nonlinear, learned analogue of a spectral method. In a spectral method you pick the basis in advance and compute the coefficients by solving equations. Here the network learns both the basis and the map from input function to coefficients, directly from data.

The implementation is in JAX/Flax, with optax for optimisation. One detail worth flagging, because it is both a clarity win and a performance win: the model is written in *outer-product* form. The branch runs once per input function, the trunk once per query point, and they combine in an `einsum`:

```
b = MLP(2 * latent, branch_hidden, branch_depth)(g_norm) # (N, 2p)
t = MLP(2 * latent, trunk_hidden, trunk_depth)(t_in)      # (M, 2p)

b_u, b_v = jnp.split(b, 2, axis=-1)
t_u, t_v = jnp.split(t, 2, axis=-1)

u = jnp.einsum("np,mp->nm", b_u, t_u) * scale + bias[0]
v = jnp.einsum("np,mp->nm", b_v, t_v) * scale + bias[1]
```

The naive alternative — flattening the data into `(function, point)` rows and evaluating the branch once per row — recomputes the branch thousands of times per function for no reason. Switching to the form above made training roughly two orders of magnitude cheaper in this experiment, and it states the DeepONet ansatz more directly.

We give u and v their own branch/trunk latent pair, since the two velocity components have quite different spatial structure.

Defining the Input Function

We need a family of lid profiles that is varied enough to be interesting but structured enough to describe. We use a four-term expansion:

$$g_{\text{raw}}(x) = a_0 + a_1 \sin(\pi x) + a_2 \sin(2\pi x) + a_3 \cos(\pi x),$$

with $a_0 \sim \mathcal{U}(0.8, 1.2)$ and $a_1, a_2, a_3 \sim \mathcal{U}(-0.35, 0.35)$, subject to a rejection rule $|a_1| + |a_2| + |a_3| \leq 0.6 a_0$. That constraint guarantees $g_{\text{raw}} \geq 0.4 a_0 > 0$, so the lid always drags in one direction and the flow stays a single primary vortex with corner eddies. Across all 70 cases, peak lid speed ranges from 0.75 to 1.74 and mean lid speed from 0.61 to 1.27. With $\nu = 0.01$ and $L = 1$, those correspond to case-specific Reynolds scales of about 75–174 (peak-speed definition) or 61–127 (mean-speed definition), even though the coefficient labelled **Re** in the nondimensional PDE remains 100 throughout.

Handling the corner singularity

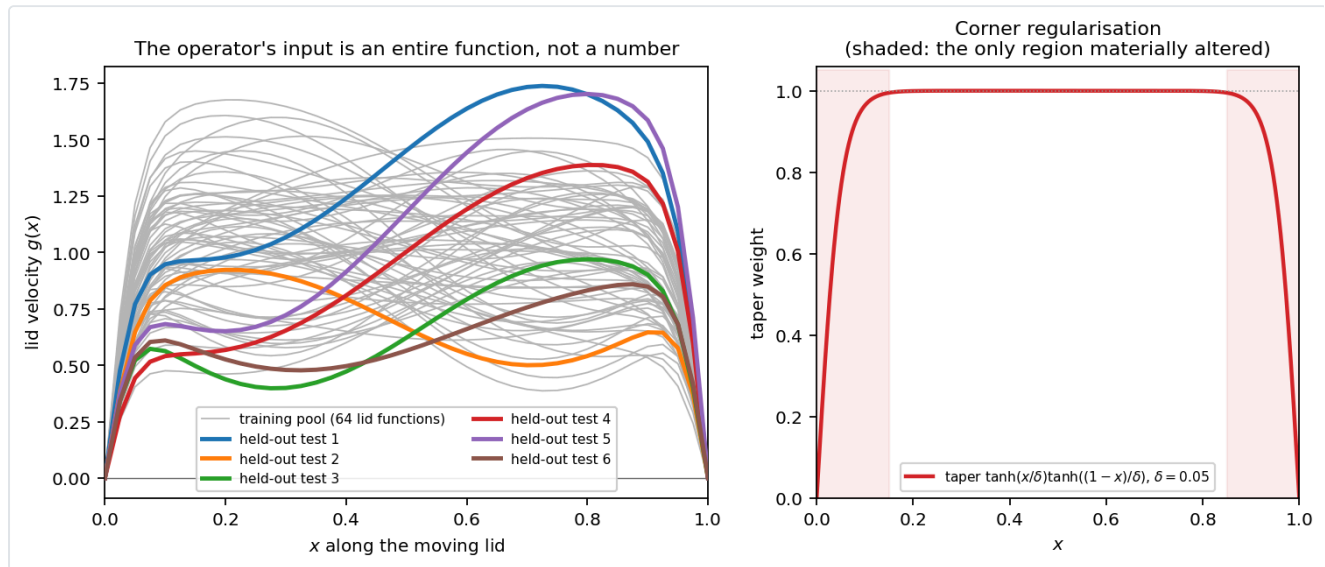
The textbook lid-driven cavity has a genuine discontinuity at the two top corners: the lid moves at $u = 1$ while the side walls are no-slip, so u jumps from 1 to 0 across zero distance. That singularity is not resolvable on any finite grid. The wall vorticity there diverges as $h \rightarrow 0$, and the resulting pollution would sit in our “ground truth” data.

We use the standard **regularised cavity** trick and multiply every profile by a smooth corner taper:

$$g(x) = g_{\text{raw}}(x) \cdot \tanh\left(\frac{x}{\delta}\right) \tanh\left(\frac{1-x}{\delta}\right), \quad \delta = 0.05.$$

Two things make this defensible rather than a fudge. First, the window exceeds 0.99 for $x \in [0.15, 0.85]$, so the shape of g is essentially untouched over the middle 70% of the lid; only a thin corner layer a few grid cells wide is smoothed. Second — and more importantly — *every* profile in the family gets the identical window. The taper is part of the operator’s input distribution, not a per-sample adjustment. The network is learning the map on the tapered family, and we test it on tapered functions too.

The right-hand panel below shows exactly how localised the taper is.



The family of input lid functions, with the corner taper

The left panel shows how DeepONet receives the problem: every grey curve is one sampled input function and each coloured curve is a held-out input. At the same time, these functions lie on a four-dimensional parameterised manifold because they are generated by (a_0, a_1, a_2, a_3) . Both descriptions matter: the model uses function samples, while the experiment covers only a restricted family.

Generating the Reference CFD Solutions

Since the training signal is supervised, the reference solutions are the weakest link in the chain: any error in them becomes the ceiling on what the operator can be shown to have learned. So it is worth being explicit about how they were produced and checked.

Why streamfunction–vorticity

We solve the steady incompressible Navier–Stokes equations in **streamfunction–vorticity** form rather than with a primitive-variable projection method. Because

$$u = \frac{\partial \psi}{\partial y}, \quad v = -\frac{\partial \psi}{\partial x},$$

the discrete velocity field is divergence-free **by construction**. There is no pressure–Poisson iteration that can be left half-converged and silently contaminate the data. Measured discrete divergence in the generated dataset is at most 4.0×10^{-15} — machine precision, not “small if the solve converged”.

The equations marched are

$$\frac{\partial \omega}{\partial t} + u \frac{\partial \omega}{\partial x} + v \frac{\partial \omega}{\partial y} = \nu \nabla^2 \omega, \quad \nabla^2 \psi = -\omega,$$

with $\nu = 1/\text{Re}$, second-order central differences on a uniform 41×41 grid, and Thom’s wall-vorticity condition. For the moving top wall,

$$\omega_{\text{wall}} = -\frac{2\psi_{\text{one node in}}}{h^2} - \frac{2g(x)}{h}.$$

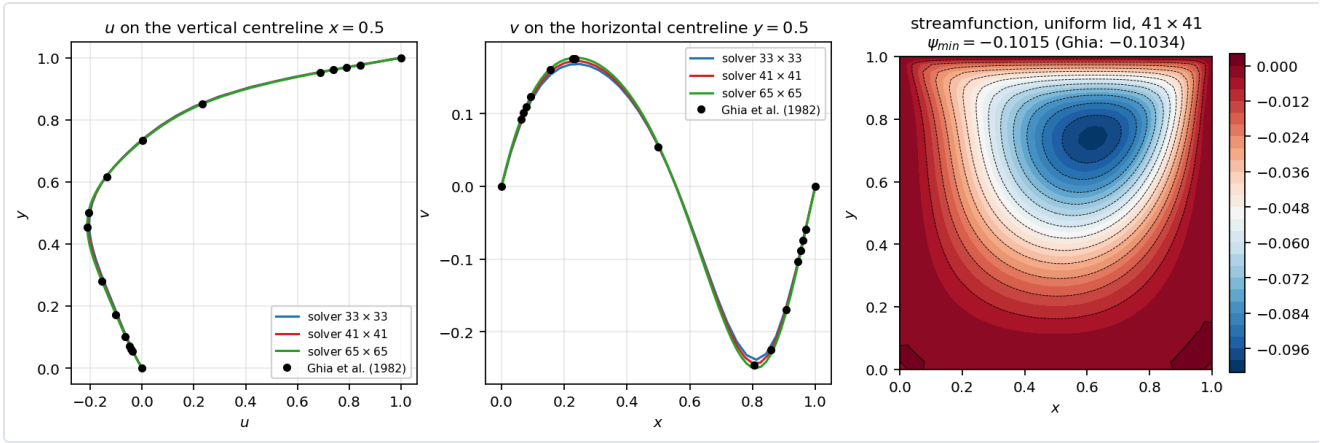
The streamfunction Poisson operator is the same matrix at every pseudo-time step, so it is LU-factorised once with `scipy.sparse.linalg.splu` and reused — an exact solve rather than a truncated iterative sweep.

Checking it, rather than trusting it

Two checks were run before any of this data was used for training.

Steadiness. The solver marches until $\|\partial \omega / \partial t\|_{\infty} < 10^{-6}$ rather than for a fixed step count. All 70 solutions in the dataset converged, taking a mean of 4876 steps, with a worst-case final residual of 9.99×10^{-7} . Nothing here is a snapshot of a still-evolving flow.

Benchmark accuracy. We ran the classical uniform-lid case ($g \equiv 1$, no taper) and compared centreline velocities against Ghia, Ghia & Shin (1982) at $\text{Re} = 100$.



Solver validation against the Ghia et al. benchmark

At the 41×41 resolution used for the dataset:

Quantity	Value
u on vertical centreline, relative L_2 vs Ghia	0.56%
v on horizontal centreline, relative L_2 vs Ghia	1.62%
max abs. deviation in u	0.0053
max abs. deviation in v	0.0034
$\psi_{\min}$	-0.1015 (Ghia: -0.1034)
max discrete $ \nabla \cdot \mathbf{u} $	3.1×10^{-15}

The three grid resolutions in the figure sit on top of the benchmark points, and $\psi_{\min}$ moves monotonically toward the reference as the grid refines (-0.1004 at 33^2 , -0.1015 at 41^2 , -0.1027 at 65^2). The primary vortex and both bottom corner eddies are resolved.

This solver is still an approximation. It uses second-order interior differences on a coarse uniform grid with a first-order wall-vorticity condition. Every “error” reported later means *disagreement with this reference*, not distance from the exact Navier–Stokes solution. The Ghia check validates one uniform, untapered case; it does not directly quantify discretisation error for the variable tapered lids. Establishing that would require finer-grid references for representative cases.

Generating all 70 solutions took 31.7 seconds. The dataset is cached under a hash of the settings that affect it, so changing a network hyperparameter never triggers a CFD rerun.

What the Training Dataset Actually Represents

It is worth being precise about what the network is being shown, because this is where the conceptual difference from a PINN lives.

Input-function pool	64 lid profiles (56 for fitting, 8 for validation)
Held-out test functions	6
Total CFD solutions	70
Grid	$41 \times 41 = 1681$ points
Supervised targets	188,272 values ($56 \times 1681 \times 2$)
Nondimensional PDE coefficient	$\text{Re} = 100$ ($\nu = 0.01$), fixed
Peak lid speed across all 70 cases	0.75 to 1.74
Corresponding peak-speed Reynolds scale	75 to 174
Primary vortex strength $\psi_{\min}$	-0.146 to -0.066

Each training example is a *pair of functions*: one lid profile and its entire corresponding velocity field. The network is not being taught the Navier–Stokes equations. It is being shown 56 input–output pairs of the solution operator and asked to interpolate between them **in function space**.

The split deserves a note, because it turns out to matter. The 6 test profiles were drawn from the same base family but subject to a **minimum-separation rule**: each is at least 12% away from every fitting profile and from every other test profile. The 8 validation profiles had no such rule — they are simply the tail of the pool. The consequence:

- validation lids sit a mean of **7.5%** from the nearest fitted lid,
- held-out test lids sit a mean of **14.8%** away.

Distance here is the symmetric normalised metric $2\|g - f\|_2 / (\|g\|_2 + \|f\|_2)$. The test set is deliberately about twice as far out. It is therefore a separated **challenge set**, not an IID sample for estimating average error under the original coefficient distribution.

The Branch and Trunk Networks

The final architecture, selected by a sweep described in the next section:

Component	Configuration
Branch input	41 lid samples $[g(x_1), \dots, g(x_{41})]$
Branch network	3 hidden layers, width 128, tanh
Trunk input	(x, y) , mapped to $[-1, 1]^2$
Trunk network	4 hidden layers, width 160, tanh
Latent basis size p	128 per velocity component
Output	$[u(x, y), v(x, y)]$
Total parameters	190,402

The sensor locations are the 41 grid x -nodes, so the branch sees the lid profile at the same resolution the solver used. Inputs are standardised per-sensor and outputs per-component, with statistics computed on the fitting split only. A $1/\sqrt{p}$ factor on the dot product keeps it $O(1)$ at initialisation. `tanh` is used throughout because the target fields are smooth.

Note what the trunk does *not* receive: it never sees g . It only sees a coordinate. All information about which cavity problem we are solving reaches the output through the 128 branch coefficients. That bottleneck is the architectural statement of “this is a map from functions to functions”.

Training the Neural Operator

The loss is plain MSE on (u, v) in normalised units. **There is no PDE residual term anywhere.** No continuity equation, no momentum equation, no boundary-condition penalty. The physics enters only through the data. This is deliberate: it isolates what operator learning is, separately from what physics-informed training is.

Final settings:

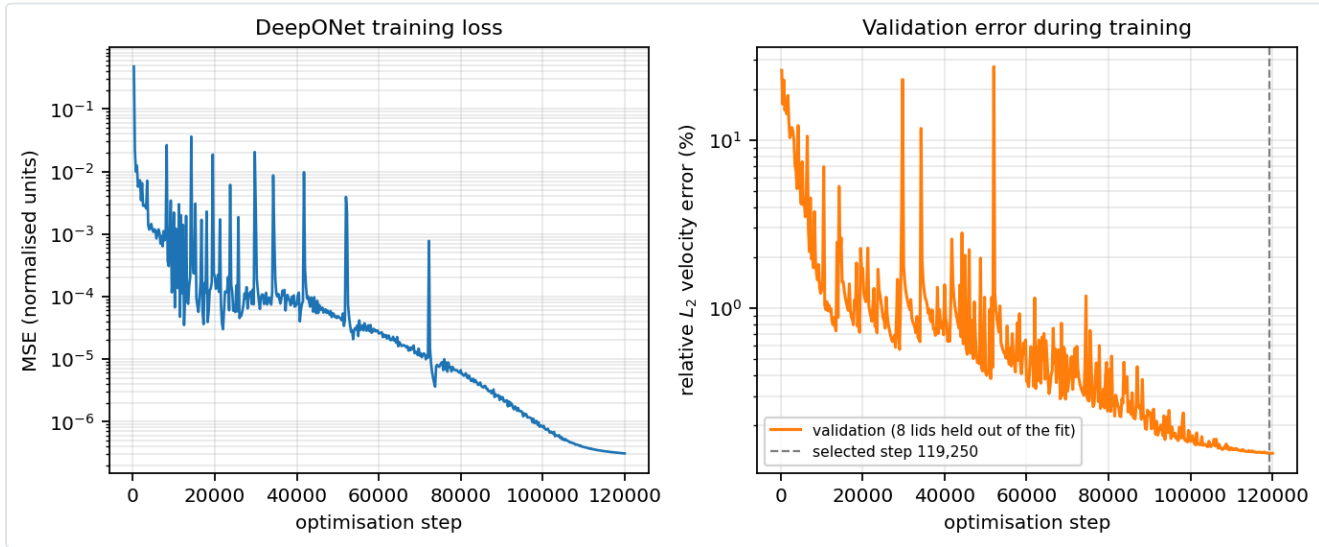
Setting	Value
Optimiser	AdamW, weight decay 10^{-4}
Learning rate	2×10^{-3} , 3% warmup then cosine decay to 10^{-5}
Gradient clipping	global norm 1.0
Steps	120,000
Batch	all 56 fitting functions $\times$ all 1681 grid points per step
Seed	7
Hardware / backend	CPU (JAX)
Wall-clock training time	35.4 minutes

Getting here took two rounds of correction, both worth recording because the first attempt looked like the sort of result one might be tempted to keep.

The first run was bad. With plain Adam at 2×10^{-3} and no clipping, training MSE fell to 1.6×10^{-5} while validation error *rose* to 5.4%, after a loss spike around epoch 700 that the run never fully recovered from. The widening training/validation gap was consistent with poor generalisation from the 42 fitting functions in that prototype.

What fixed it: gradient clipping (removing the spikes), a small decoupled weight decay, more input functions ($48 \rightarrow 64$), and a longer cosine schedule. A 10-configuration sweep followed by two longer candidate runs selected the final model, scored **only on the eight validation functions**. The 6 test functions were not evaluated until the final model was fixed. The sweep results are in `outputs/data/sweep_results.json`.

Because the same small validation set was checked every 250 steps and reused for architecture selection, its final 0.137% error may itself be optimistic. The separated test set remains untouched by that selection, but it answers a different, harder question.



DeepONet training loss and validation error

The left panel shows the training MSE reaching 3.1×10^{-7} . The spikes in the first half are real and worth not hiding: at this learning rate the optimiser periodically takes a bad step, and gradient clipping plus the cosine decay is what tames them. After roughly step 80,000 the run is monotone.

The right panel is the one that matters. Validation error falls alongside training error and keeps falling to the end — the best model is at step 119,250, essentially the final one. Unlike the failed prototype, this run shows no late training/validation divergence. That is encouraging evidence, although it is not by itself proof against memorisation or validation overfitting.

Final validation error: **0.137%** relative L_2 on the combined velocity field.

Testing on Unseen Lid Functions

Now the actual question. Six lid profiles the network has never seen, each at least 12% away under the symmetric normalised L_2 metric from anything it was trained on. For each: solve the CFD reference, run one forward pass of the DeepONet, compare.

The coefficient vectors (a_0, a_1, a_2, a_3) of the held-out test functions, recorded for reproducibility:

Test	a_0	a_1	a_2	a_3	Distance to nearest fitted lid
1	1.1813	0.2703	−0.2700	−0.1294	12.0%
2	0.8293	−0.1646	0.2188	−0.0213	12.9%
3	0.8210	−0.1984	−0.2589	−0.0246	15.2%
4	0.9426	0.0608	−0.1534	−0.3263	13.6%
5	1.1757	0.0028	−0.2824	−0.3179	18.9%
6	0.8425	−0.2774	−0.1056	−0.0650	16.4%

Results

Test	rel. L_2 (combined)	rel. $L_2 u$	rel. $L_2 v$	MAE u	max abs. err u	centreline u err
1	1.53%	1.49%	1.63%	0.0028	0.0355	1.36%
2	0.27%	0.24%	0.33%	0.0003	0.0027	0.19%
3	1.78%	1.76%	1.84%	0.0016	0.0172	2.02%
4	0.68%	0.66%	0.72%	0.0008	0.0119	0.39%
5	1.76%	1.86%	1.49%	0.0024	0.0370	0.70%
6	1.65%	1.63%	1.71%	0.0012	0.0165	1.81%

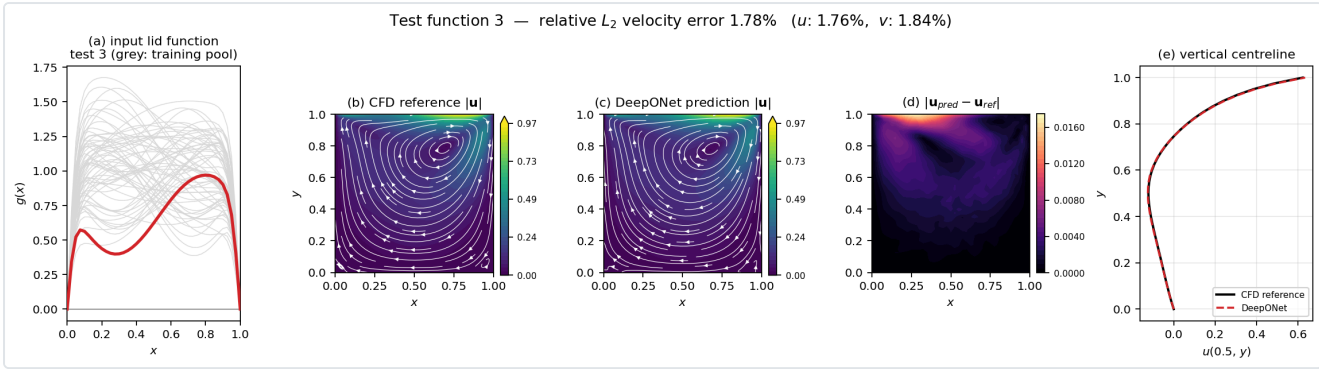
Mean relative L_2 velocity error across the six unseen lid functions: 1.28%. Best 0.27%, worst 1.78%. The u and v components are essentially equally well predicted (1.27% and 1.29% on average).

Because this is a deliberately separated six-case challenge set, the mean is uncertain and should not be read as population performance. The median is 1.59%; a descriptive 95% t interval for the mean is 0.60%–1.95%. More random test functions and repeated training seeds would be needed for a tight generalisation estimate.

For Test Function 2, the relative L_2 velocity error was 0.27%. For Test Function 3 — the worst case — it was 1.78%.

Input Function → Output Flow Field

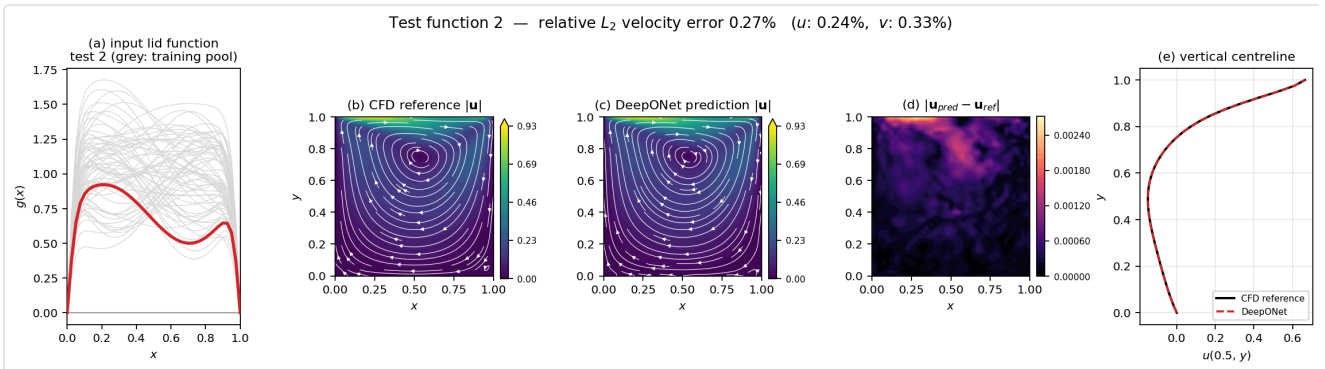
Here is a single test case in full. Panel (a) is the input function; (b) is the CFD reference; (c) is the DeepONet’s prediction; (d) is the pointwise error magnitude; (e) compares the vertical-centreline horizontal velocity.



Test function 3: full comparison

This is the worst of the six cases at 1.78%, and the predicted flow is still visually indistinguishable from the reference: same primary vortex, same off-centre vortex position, same corner behaviour, centreline profiles lying on top of each other. The error field (d) shows where the disagreement actually lives — a thin band immediately under the moving lid, peaking around 0.017, where velocity gradients are steepest.

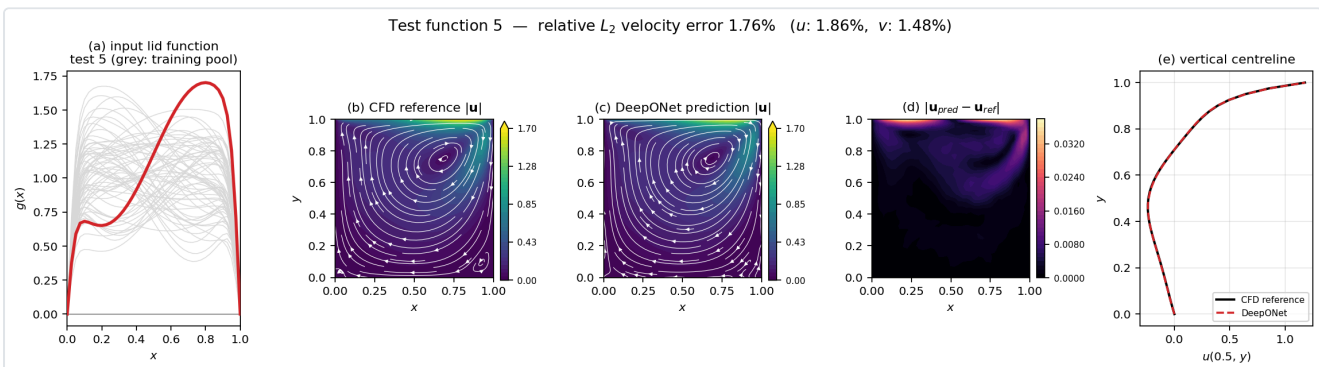
Compare with the best case:



Test function 2: full comparison

Test 2 has a lower-amplitude, flatter lid profile, and the error drops by almost an order of magnitude — peak error around 0.0025, distributed as low-level noise rather than concentrated structure.

And a high-amplitude case, where the lid peaks at about 1.7:



Test function 5: full comparison

Here the error concentrates in the lid boundary layer again, and the bottom-right corner eddy is rendered slightly less crisply than in the reference. Faster lids mean thinner boundary layers, and the boundary layer is consistently the hardest part of the field.

The remaining test-case summaries are saved alongside these figures in `outputs/figures/`.

CFD Reference vs Neural Operator Prediction

Something is worth pausing on. To produce the reference field, the solver marched roughly 4900 pseudo-time steps, solving a sparse linear system at each one. To produce the prediction, the network did one forward pass.

Measured on the same machine (medians over repeated runs; wall-clock timings depend on hardware and load, so treat these as an order of magnitude rather than a benchmark):

	Time per velocity field
CFD solve (41×41 , to steady state)	≈ 498 ms
DeepONet forward pass	≈ 1.9 ms

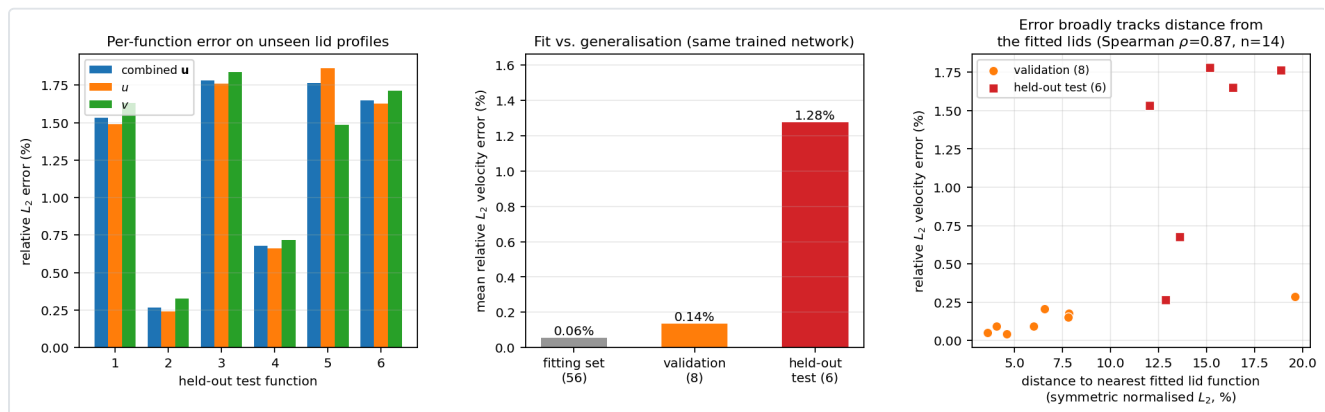
That is roughly a **260 $\times$ speedup** at inference, for about 1% disagreement on the separated test cases.

The marginal deployment accounting includes 31.7 seconds of CFD to build the dataset plus 35.4 minutes for the selected training run. Dividing that overhead by the per-solve saving gives about **4,300** new lid profiles before that final run breaks even against simply running the solver each time.

That is not the total research cost. The recorded 10-run sweep and two final candidate runs, together with dataset generation, total at least 3.0 hours and raise the recorded break-even lower bound to about **22,000 solves**. The earlier failed prototype was not timed in machine-readable form and is excluded from that larger figure.

That is the real economic shape of operator learning, and it is not “neural networks beat CFD”. It is: *if you need the same family of problems solved over and over — for optimisation, control, uncertainty quantification, or real-time inference — you can pay a large one-off cost to make each subsequent solve nearly free*. If you need one flow field once, run the solver.

Error Analysis



Error summary across test functions

The left panel shows per-function errors broken into components. The middle panel is the one to read carefully:

Set	Functions	Mean rel. L_2 velocity error
Fitting set	56	0.057%
Validation	8	0.137%
Held-out test	6	1.278%

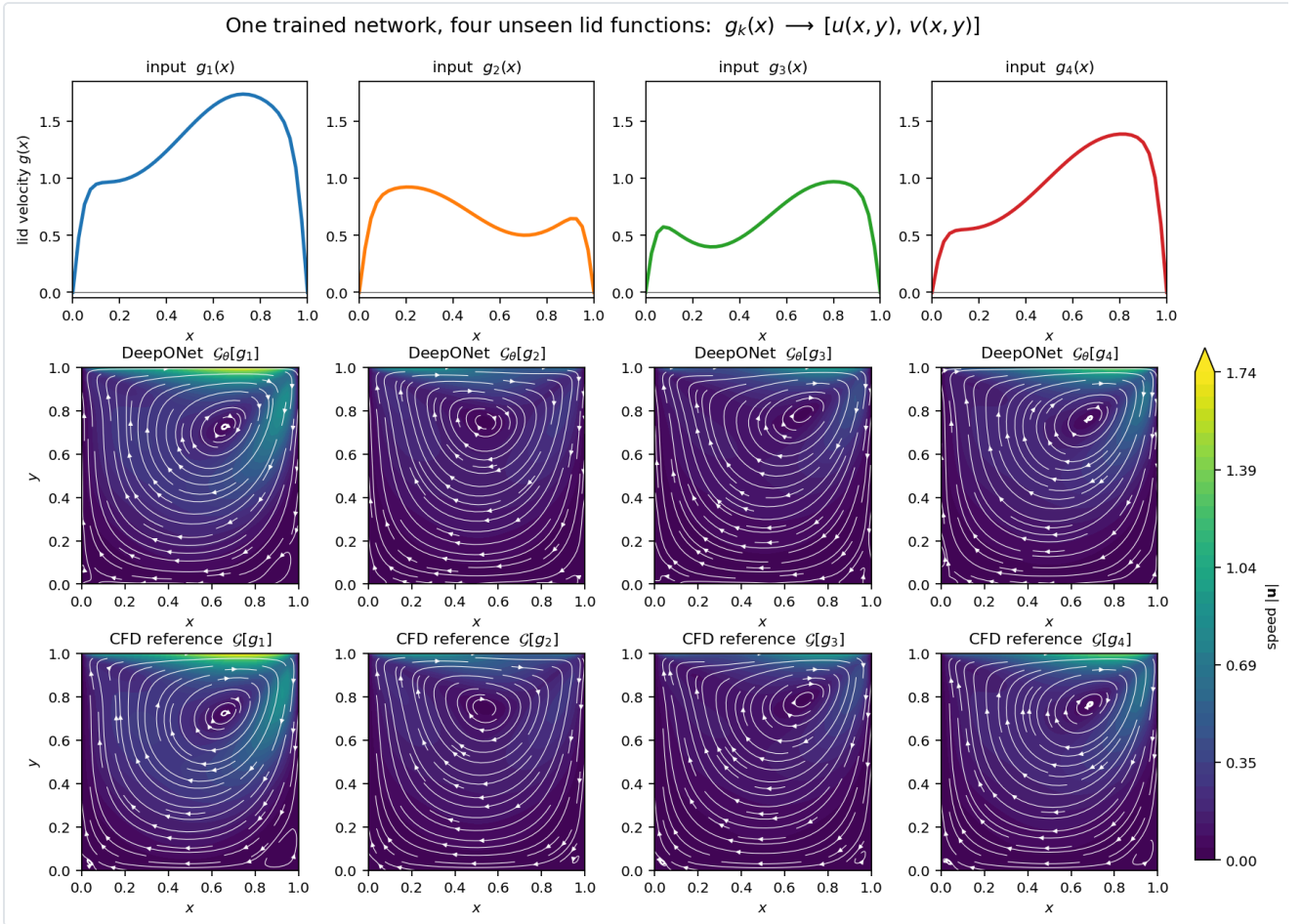
The network reproduces the functions it was fitted on to about 0.06%, nearby unseen functions to about 0.14%, and the deliberately-separated test functions to about 1.3%. Error grows by roughly an order of magnitude from the fitting and validation sets to the separated challenge set. Because set membership and distance move together, this comparison alone does not identify distance as the cause.

The right panel tests whether that is really a distance effect, by plotting each unseen function's error against its distance from the nearest fitted lid. The combined rank correlation is strong — Spearman $\rho = 0.87$ over the 14 unseen functions ($p = 5.7 \times 10^{-5}$) — but it partly reflects combining a nearby validation group with a deliberately distant test group. Within the six test cases alone, $\rho = 0.66$ and $p = 0.156$. Test 2 sits 12.9% away and is the *most* accurate case at 0.27%; one validation function sits 19.6% away yet achieves 0.30%. Distance is a plausible contributor, but this small grouped sample does not establish a general distance–error law.

Error is also structured spatially rather than uniform. Across all six test cases the largest absolute errors sit in the lid boundary layer, with a maximum of 0.037 in u against peak speeds of order 1.0 to 1.7. The interior, where the flow is smooth, is predicted far more accurately than the thin shear layer at the top.

One Network, Multiple Input Functions

This is the figure that makes the operator claim concrete.



One network, four unseen lid functions

Four different unseen lid functions go in on the top row. The middle row is what a **single trained network** produces for each — no retraining, no fine-tuning, no optimisation of any kind between columns. Just four forward passes. The bottom row is the corresponding CFD reference.

$$g_1(x) \rightarrow \mathbf{u}_1(x, y), \quad g_2(x) \rightarrow \mathbf{u}_2(x, y), \quad g_3(x) \rightarrow \mathbf{u}_3(x, y), \quad g_4(x) \rightarrow \mathbf{u}_4(x, y).$$

Look at what varies. The lids peaked toward the right (g_1, g_4) push the primary vortex centre right, to around $x \approx 0.67$. The flatter, lower-amplitude g_2 produces a weaker, more centred vortex at $x \approx 0.55$. The network is not merely rescaling one memorised flow pattern by the lid amplitude — it is responding to the *shape* of the input function, and the predicted vortex positions track the references.

That is the payoff of the whole setup. One set of 190,402 weights encodes an approximation to the solution operator across a family of cavity problems.

PINN vs Neural Operator Revisited

We can now state the contrast concretely, in terms of what was actually built.

A PINN, as in the earlier article, represents one solution to one fixed problem:

$$(x, y) \longrightarrow [u(x, y), v(x, y), p(x, y)].$$

The network’s input is a coordinate. Its weights *are* the solution to one particular cavity problem, found by minimising PDE and boundary-condition residuals. Change the lid velocity and those weights are wrong; you optimise again from scratch.

A **neural operator**, as built here, maps between problems and their solutions:

$$g(x) \longrightarrow [u(x, y), v(x, y)].$$

The network’s input is a function. Its weights encode an approximation to $\mathcal{G}$ across a family of cavity problems. A new within-family lid profile is a **forward pass**, not a new optimisation — under 2 ms here rather than a fresh training run.

The compact version:

A PINN learns a solution; a neural operator learns a mapping between problems and solutions.

That slogan is a conceptual simplification, not a definition, and it is worth qualifying immediately:

- **PINNs can be parameterised.** Feed a PINN extra inputs — Reynolds number, or coefficients describing the boundary condition — and it learns a family too. The boundary between “parameterised PINN” and “neural operator” is genuinely blurry.
- **Neural operators need not be trained from CFD data.** Ours was, because that is the cleanest demonstration. But operators can be trained from PDE residuals instead, with no reference solutions at all.
- **“Learns a mapping” overstates what was verified.** What this experiment demonstrates is a good approximation over the *distribution its training data came from*. That is much weaker than learning $\mathcal{G}$.

The useful distinction is not architectural but about *what varies across training examples*. In a PINN, nothing varies — there is one problem. Here, an entire input function varies. That is the axis that matters.

Where Did the Physics Go?

This is the question the experiment raises most sharply. We removed every physics term from the loss. So in what sense does the trained network know any fluid dynamics?

The answer is: **only what it absorbed from labeled solution fields, and we can measure the result.** The script `src/physics_check.py` checks boundary reproduction and discrete incompressibility.

The results are striking, and they cut both ways.

What it reproduced well. No symbolic boundary equation or boundary penalty was supplied, but boundary grid points and their CFD velocity labels were part of every full-grid supervised training step. Across the six test cases, the maximum predicted speed on the side and bottom walls averages **0.0009** — essentially

zero against speeds of order 1. The model also reproduces the lid profile to a mean maximum error of **0.0202**. This is accurate interpolation of boundary values seen repeatedly in the labeled fields, not evidence that the network independently discovered the boundary conditions.

What it did not learn. Incompressibility:

Quantity	Mean across six cases
Reference max $ \nabla \cdot \mathbf{u} $	2.9×10^{-15}
DeepONet max $ \nabla \cdot \mathbf{u} $	4.3×10^{-2}
DeepONet RMS divergence	7.7×10^{-3}

The maximum differs by about thirteen orders of magnitude. Normalised by the mean peak-speed scale, the DeepONet’s mean maximum divergence is **3.37%** and its mean RMS divergence is **0.61%**. Every reference field the network trained on was divergence-free to machine precision. The network’s own outputs are not divergence-free in the discrete sense used here.

This is the honest answer to “where did the physics go?”. The fields *look* right, and they are accurate to $\sim 1\%$ in L_2 , but the network reproduces the **appearance** of the solutions without inheriting the **structure** that generated them. It learned to interpolate in a space of divergence-free fields without learning that divergence-free is a constraint. Nothing in MSE training ever asked it to.

That distinction — accurate in norm, wrong in structure — is exactly the gap that physics-informed methods exist to close, and it is why “the network learned the physics” is a claim to be suspicious of when nothing enforced it.

Physics-Informed Neural Operators

The two ideas in this series are not rivals; they compose.

A **physics-informed neural operator** keeps the branch/trunk architecture from this experiment but adds PDE residual terms to the loss, evaluated by differentiating the network output with respect to its trunk inputs. The trunk’s smooth `tanh` layers make $\partial u / \partial x$, $\partial u / \partial y$ and so on available analytically through autodiff — and in JAX that is `jax.grad` on the trunk coordinates.

The loss then becomes something like

$$\mathcal{L} = \underbrace{\|\mathcal{G}_\theta[g] - \mathbf{u}_{\text{ref}}\|^2}_{\text{data}} + \lambda_1 \underbrace{\|\nabla \cdot \mathcal{G}_\theta[g]\|^2}_{\text{continuity}} + \lambda_2 \underbrace{\|\mathcal{R}_{\text{NS}}[\mathcal{G}_\theta[g]]\|^2}_{\text{momentum}}.$$

Given the divergence measurement above, the continuity term alone would target a real, quantified defect in this model. The attractions are concrete: fewer reference solutions needed (the residual supplies signal where data does not), physically consistent outputs, and often better behaviour away from the training distribution.

The costs are equally real: more expensive training steps, additional loss weights λ_i to balance, and the stiff-gradient pathologies that make PINN training finicky in the first place — now inherited by the operator.

This was not attempted here, deliberately. Mixing supervised operator learning with physics-informed training would have made it impossible to say which mechanism produced which behaviour. The clean claim of this experiment is narrower and more interpretable: *supervised operator learning alone gets about 1% disagreement on this separated challenge set, and does not get incompressibility.*

Limitations of This Experiment

Stating these precisely is what keeps the demonstration honest.

The reference solver is approximate. Second-order interior differences, uniform 41×41 grid, first-order Thom wall vorticity. It matches the Ghia benchmark to 0.56% (u) and 1.62% (v). Every reported “error” is disagreement with *this*, not with exact Navier–Stokes. The benchmark is a uniform untapered case; solver error for the variable tapered lids was not measured against a finer reference and cannot be inferred directly from those two percentages.

The learned operator is not a Navier–Stokes solver. It has no notion of the equations. Its mean maximum discrete divergence is 3.37% of the mean peak-speed scale (mean RMS: 0.61%). It cannot be refined by taking smaller time steps, and it has no error estimator — nothing in its output signals when it is wrong.

Generalisation is restricted to one parameterised family. The 4-coefficient family with fixed taper and fixed $\nu = 0.01$ is the entire world this network knows. A lid profile with a sharp step, a different taper width, negative regions, or a different Reynolds number is outside its training distribution, and there is no evidence here about how it would behave. The model was not compared with a coefficient-input MLP, POD model, nearest-neighbour interpolator, or FNO, so no claim of architectural superiority is supported.

The sample sizes are small. 56 fitting functions, 8 validation, 6 separated test cases, and one training seed. The 1.28% mean has a descriptive 95% interval of 0.60%–1.95%, and the distance relationship is confounded with split design.

The problem is easy by CFD standards. $Re = 100$ is steady, laminar, single-vortex flow. Nothing here speaks to turbulence, unsteadiness, or the Re range where cavity flow becomes genuinely hard.

Only one architecture was tried. No FNO comparison, no ablation of the branch/trunk split. We do not claim DeepONet is best for this — only that it makes the concept legible.

The physical velocity scale varies. The PDE coefficient called Re is fixed at 100, but lid amplitude varies enough that Reynolds scales based on actual peak or mean lid speed span approximately 75–174 or 61–127, respectively.

Three things this experiment does **not** support, and which the results should not be stretched to imply: that the learned operator is an exact Navier–Stokes solver; that it will generalise to arbitrary lid functions; or that neural operators generally replace CFD. It also should not be read as saying that neural operators are necessarily trained from supervised CFD data — this one was, by choice.

Final Intuition

Start where we started. The integration operator

$$f(x) \xrightarrow{\mathcal{I}} u(x)$$

takes a function and returns a function. We can write it down, so it feels unremarkable.

The Navier–Stokes solution operator

$$g(x) \xrightarrow{\mathcal{G}} [u(x, y), v(x, y)]$$

does the same thing — takes a function (the lid profile), returns functions (the velocity field). The difference is that we cannot write $\mathcal{G}$ down. We can only evaluate it, expensively, one input at a time, by running a solver.

What this experiment shows is that $\mathcal{G}$ can be *approximated by learning from examples of its input–output behaviour*. Show a DeepONet 56 (lid profile, velocity field) pairs with fixed $\nu = 0.01$, and it recovers an approximation accurate to **1.28% on average** across six deliberately separated lid profiles it never saw — in under **2 ms** per field instead of roughly half a second. The median is 1.59%, and the small-sample interval for the mean is wide (0.60%–1.95%).

The mental shift is the point. A PINN asks: *what field satisfies this PDE?* Its answer is a set of weights encoding one solution. A neural operator asks: *what is the pattern relating this family of problems to their solutions?* Its answer is a set of weights encoding an approximate map.

The trunk network learned a basis of 128 spatial functions over the cavity. The branch network learned to read a lid profile and decide how much of each basis function it calls for. Between them they compress a family of PDE solutions into 190,402 numbers.

But hold that alongside the divergence measurement. This network produces fields that are accurate to about 1% and yet has a mean maximum discrete divergence equal to 3.37% of the characteristic speed scale. It captured the appearance of the solutions without the structure underneath. A learned operator is a fast, powerful interpolator over the distribution it was shown — and it is precisely that, no more. Knowing which of those two things you are holding is the difference between using it well and being misled by it.

Reproducing this

```
cd neural_operator_cavity
python -m venv .venv && source .venv/bin/activate
pip install -r requirements.txt
python run_all.py
```